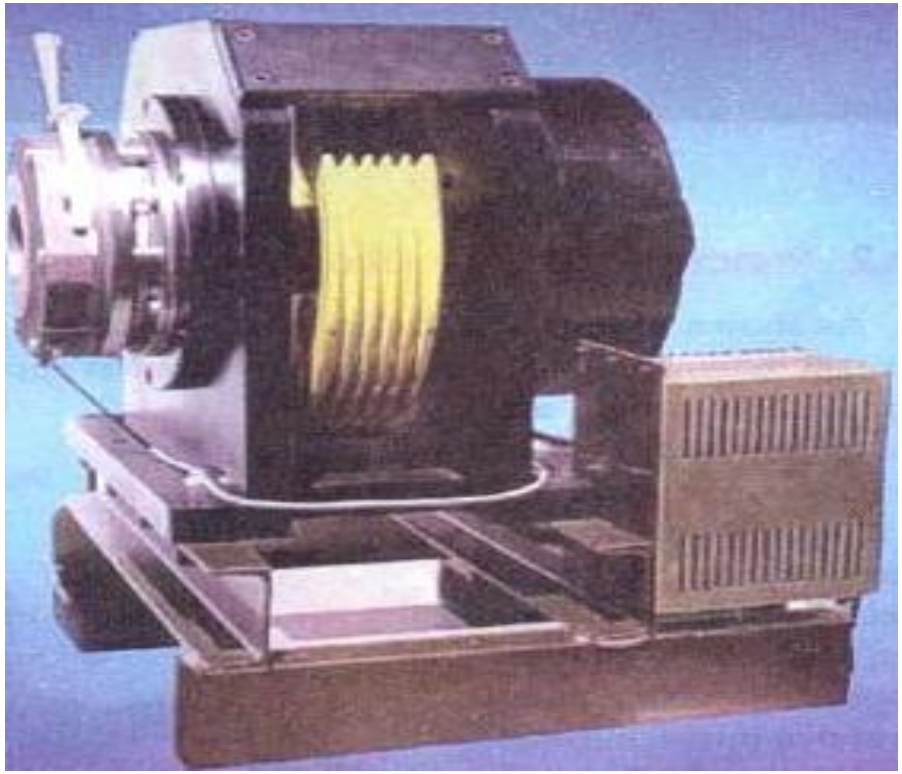
EEE 3507

Synchronous Motors

Synchronous Motors

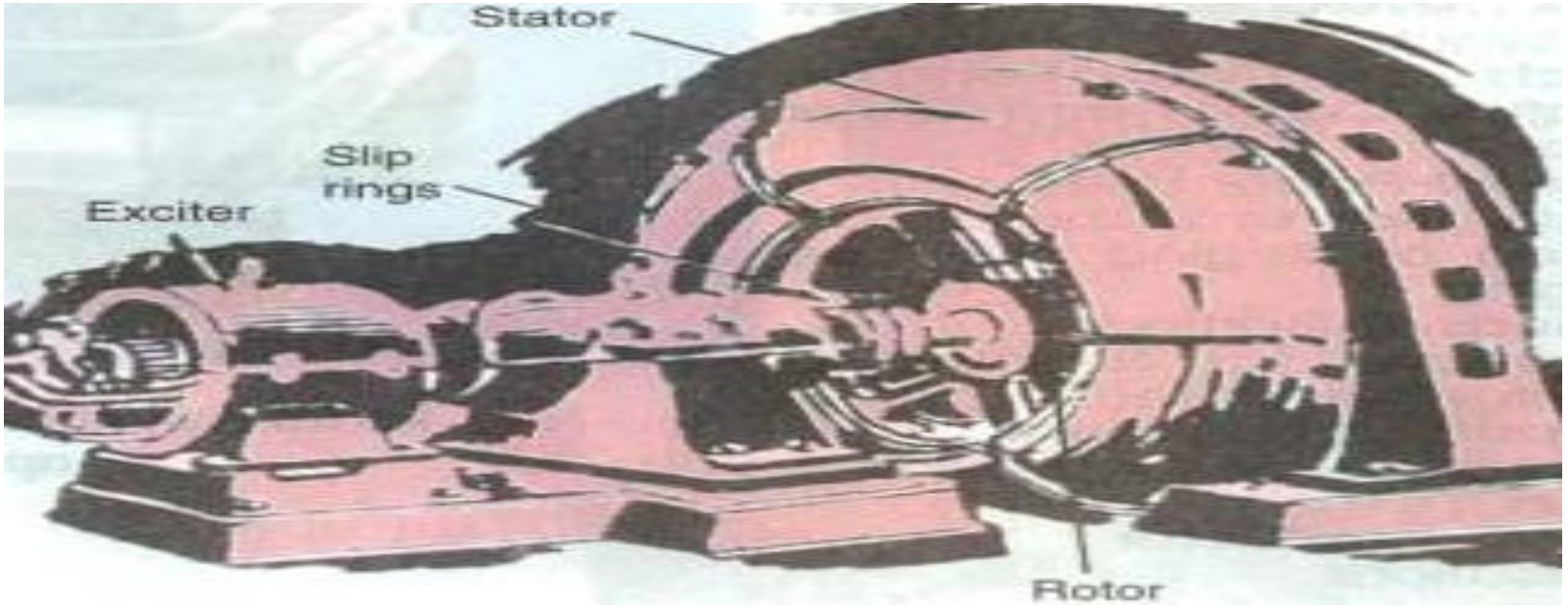
- An alternator may operate as synchronous motor **like DC m/c**
- Synchronous motor (SM) is very similar to a synchronous gen.





Synchronous Motors (Cont.)

- Rated 15 MW~ 150kW (150~1800 rpm)..
- It runs either at synchronous speed* Ns (120f/P) or not at all but N can be changed from f changing (f - converting)..
- It is not inherently self starting…
- Can be operated wide range of p.f. (lag-unity-lead) by field..



- **Synchronous speed:** The speed at which the rotating field moves around the stator is called synchronous speed, Ns = 120f/P, usually measured in rpm.

Applications of Synchronous Motors

1. Synchronous motors are particularly attractive for constant speed & low speeds (< 300 r.p.m.) because the power factor can always be adjusted to unity and efficiency is higher
2. Overexcited synchronous motors can be used to improve the pf of system where lagging loads (IM, light, welders.. etc.)
3. They are used to improve the voltage regulation of transmission line (SM with field regulator, rise & fall controlled)
4. High-power electronic converters generating very low frequencies enable us to run SM at ultra-low speeds, Large motors in the 10 MW range are use in industry and variable-speed ball mills etc..

Construction of SM

11.1 Construction

A synchronous motor is a machine that operates at synchronous speed and converts electrical energy into mechanical energy. It is fundamentally an alternator operated as a motor. Like an alternator, a synchronous motor has the following two parts:

(i) a stator which houses 3-phase armature winding in the slots of the stator core and receives power from a 3-phase supply [See (Fig. (11.1)].

(ii) a rotor that has a set of salient poles excited by direct current to form alternate N and S poles. The exciting coils are connected in series to two slip rings and direct current is fed into the winding from an external exciter mounted on the rotor shaft.

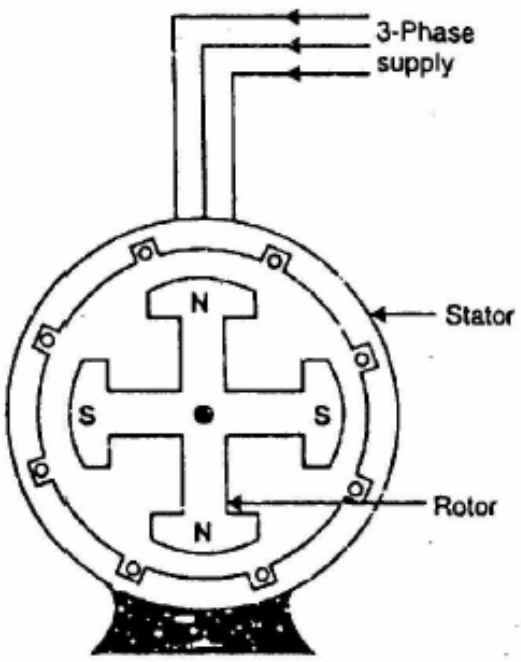
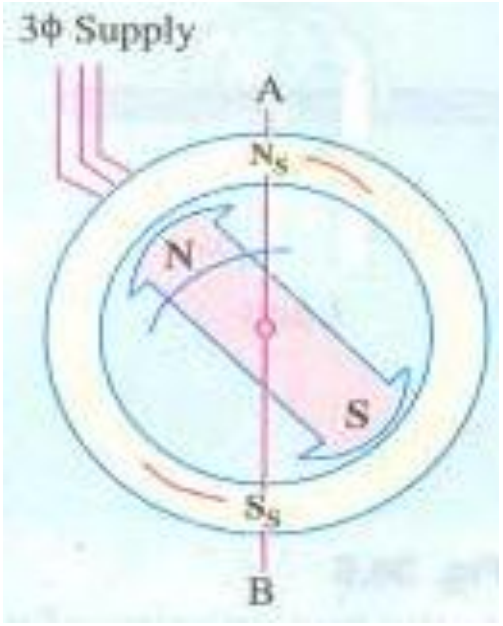


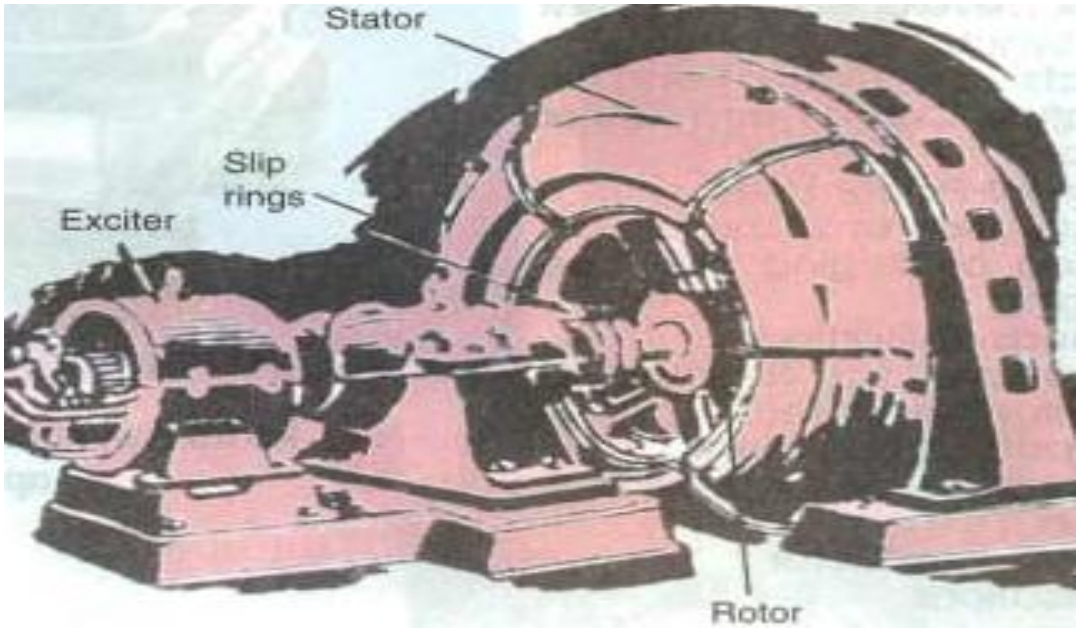
Fig.(11.1)

The stator is wound for the same number of poles as the rotor poles. As in the case of an induction motor, the number of poles determines the synchronous speed of the motor:

Operating Principle

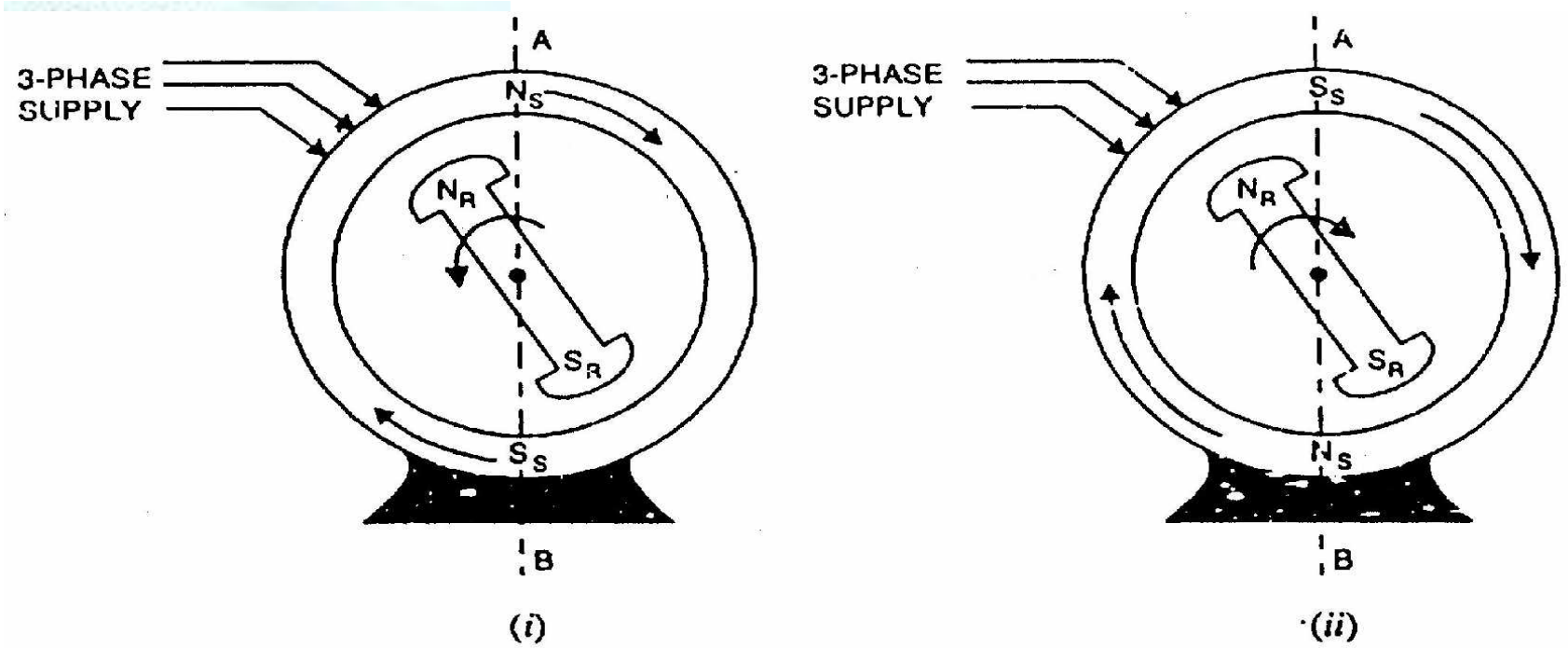
- Consider a 3-phase SM having two rotor poles N_R and S_R
- The stator is designed for two poles N_S and S_S
- DC applied to the rotor and a 3-phase voltage applied stator
- The **stator winding produces a rotating field** which revolves round the stator at synchronous speed Ns(=120 f/P)
- A pair of revolving armature poles (i.e., Ns - Ss) and a pair of stationary rotor poles (i.e., N_R - S_R)





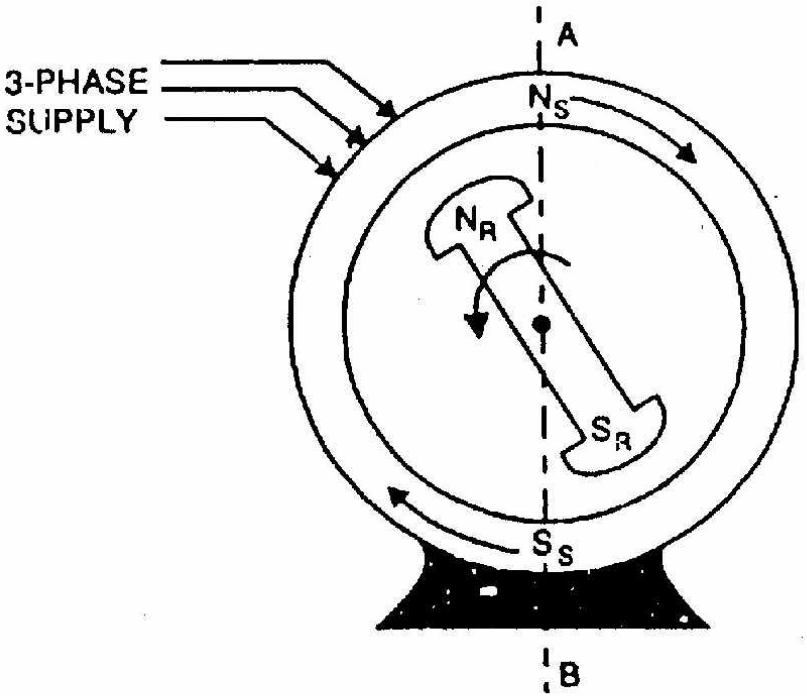
Operating Principle (Cont.)

- Suppose at any instant, the stator poles are at positions Fig. N_S and N_R repel each other and so do the poles Ss and S_R. The rotor tends to move in the ccw direction
- After a period of half-cycle, the polarities of the stator poles are reversed but not the rotor poles. Now Ss and N_R attract each other and so do Ns and S_R.

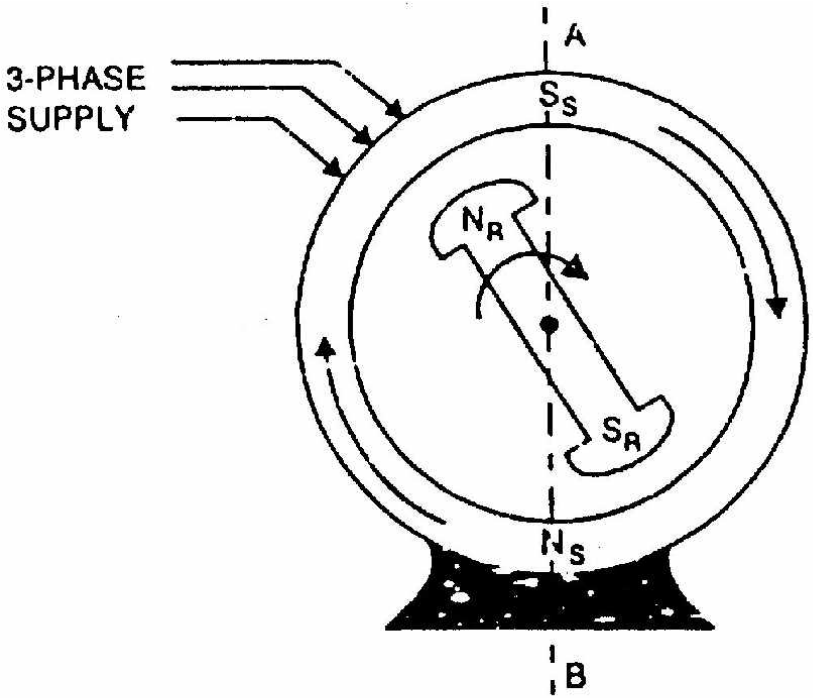


Operating Principle (Cont.)

➢Therefore, the rotor tends to move in the clockwise direction. Since the stator poles change their polarities rapidly, they tend to pull the rotor first in one direction and then after a period of half-cycle in the other. Due to high inertia of the rotor, the motor fails to start. Thus, **SM has no self-starting torque**



(i)

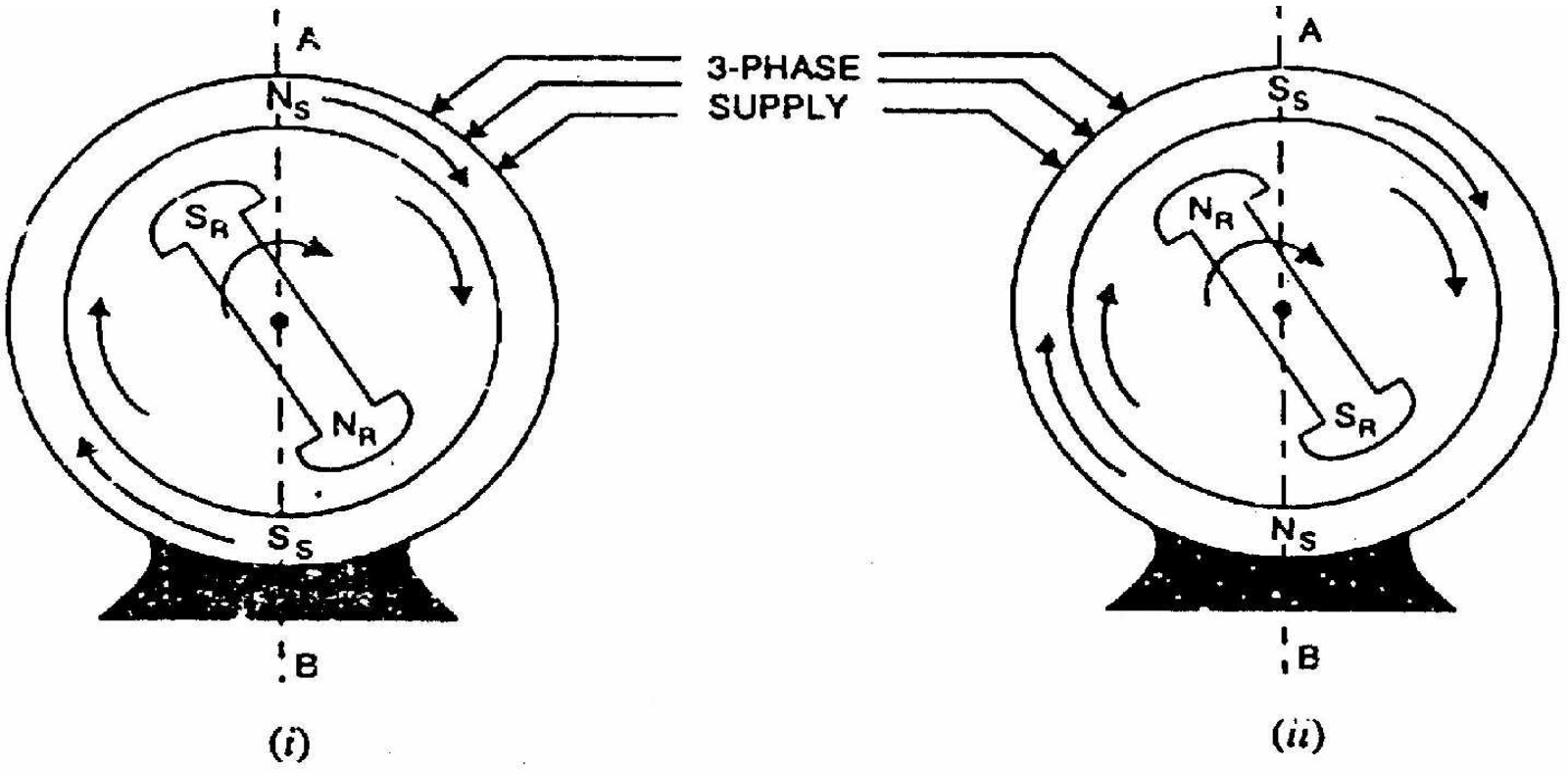


(ii)

Operating Principle (cont.)

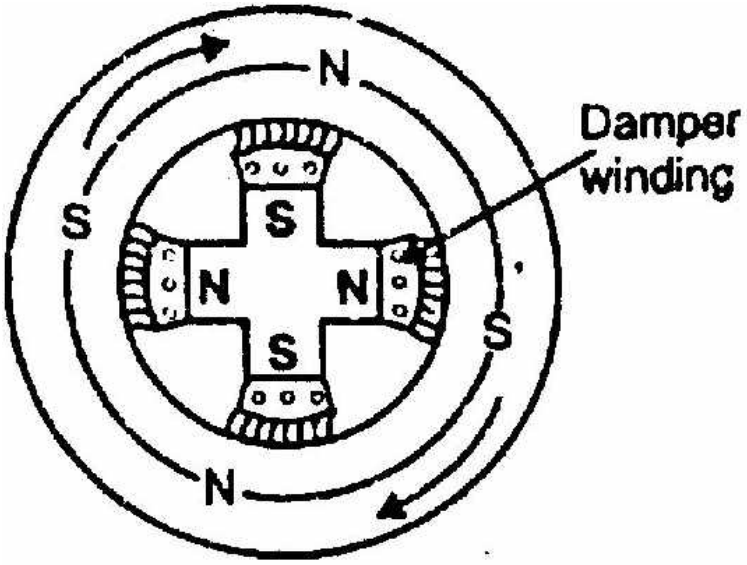
➢**How to get continuous unidirectional torque?** If the rotor poles are rotated by some external means at such a speed that they interchange their positions along the stator poles, then rotor will experience a continuous unidirectional torque

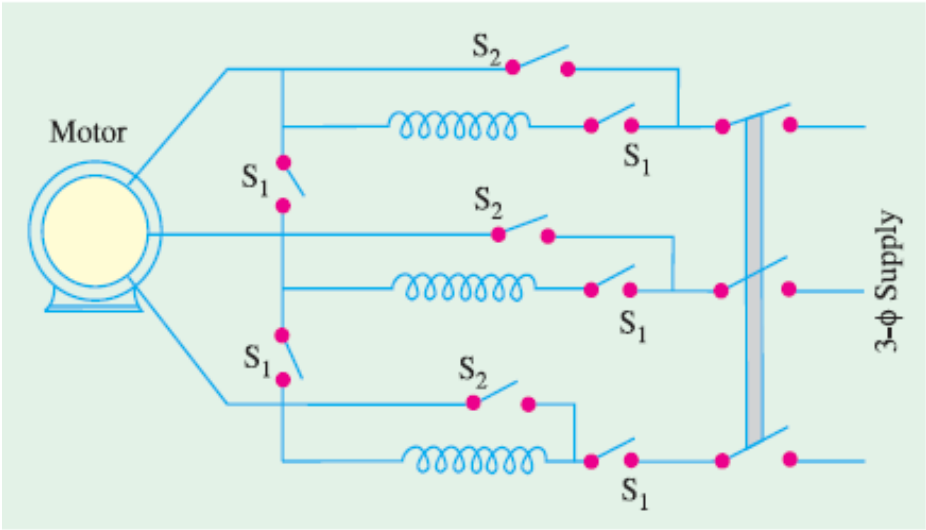
➢N_R-Ss locked and vice versa both rotating synchronism at synchronous speed of the supply voltage.



Methods of Starting

- The rotor has to be speed up Ns/near Ns by arrangements…
- N_R-Ss and vice versa both interlocked for synchronism
- In reality rotor brought up to Ns by a) A shunt motor b) Exciter as a DC motor c) IM (less poles) & d) Damper winding as a IM



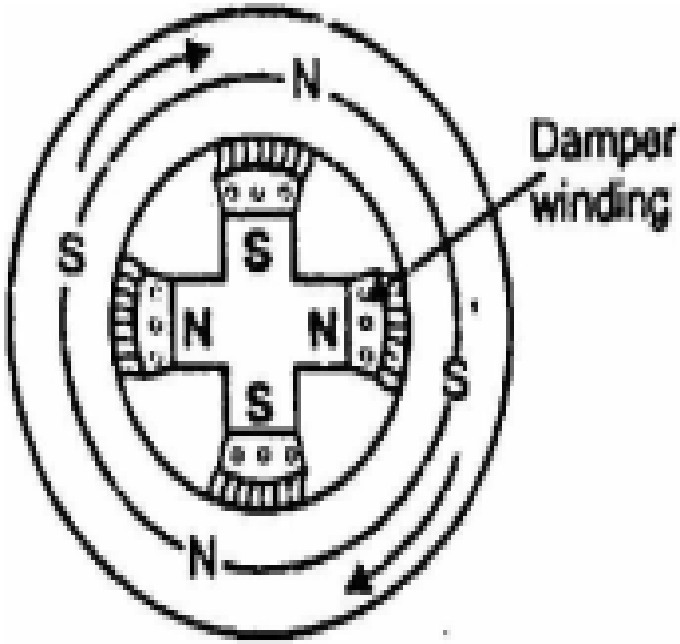


Making SM Self-Starting

A synchronous motor cannot start by itself., If stator is feed & rotor excitation is given motor develop no torque at standstill. It will develop running torque once it has been brought up to Ns. In order to make the motor self-starting, There are several methods of accomplishing this.

a) Induction start
b) Auxiliary devices

Induction start: All most all SM are equipped with squirrel cage winding (also called damper winding) is provided on the rotor. The damper winding consists of copper bars embedded in the pole face. The bars are short-circuited at both ends. Act as a IM during starting period



Making SM Self-Starting (Cont.)

Auxiliary devices:

(i) A direct coupled: DC/Induction Motor can be used rating is about 10% of SM, Pole less (min. 2)

(ii) SM as Alternator

Synchronized with supply

Now remove IM, will run as SM

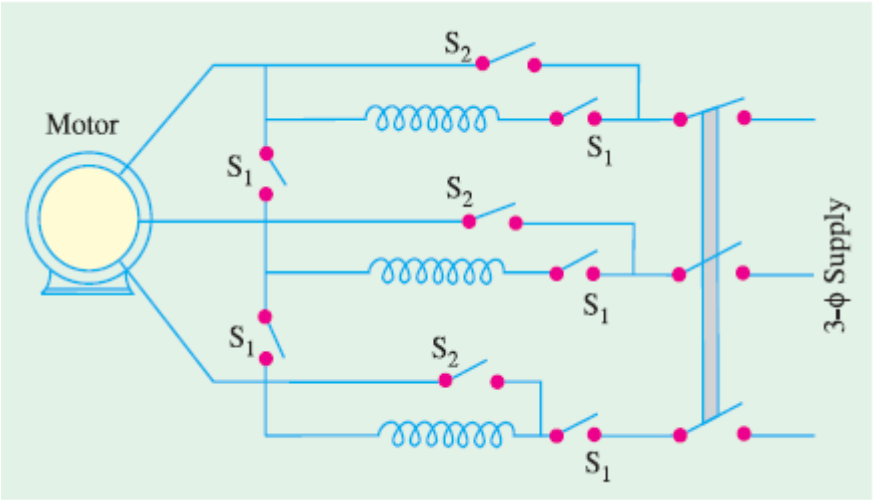
(iii) Without Synchronizing:

Need to speed up by IM or DC motor near at Ns

When Ns is reached, Stator is feed and rotor is excitated…

SM start……

(iv) Using the field exciter generator as a DC motor



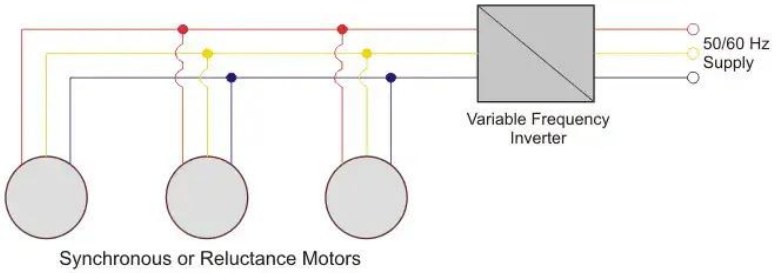
Comparison of SM & Induction Motors

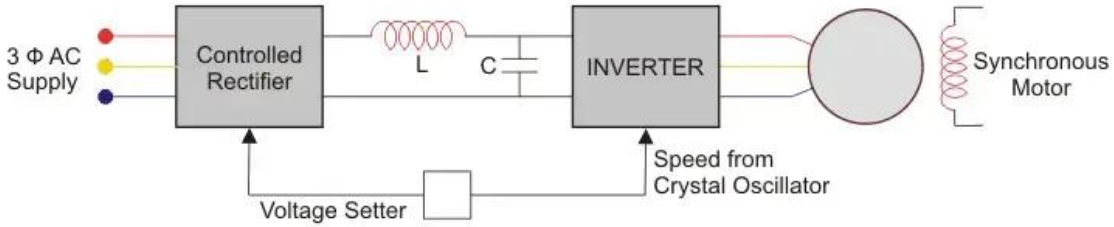
Particular	Synchronous Motor	3-phase Induction Motor
Speed	Remains constant (i.e., N_s) from no-load to full-load.	Decreases with load.
Power factor	Can be made to operate from lagging to leading power factor.	Operates at lagging power factor.
Excitation	Requires d.c. excitation at the rotor.	No excitation for the rotor.
Economy	Economical fcr speeds below 300 r.p.m.	Economical for speeds above 600 r.p.m.
Self-starting	No self-starting torque. Auxiliary means have to be provided for starting.	Self-starting
Construction	Complicated	Simple
Starting torque	More	less

A combination of rectifiers and inverters can be used to control the speed of synchronous motors. There are two main methods for this:

Inverter Fed Open Loop Synchronous Motor Drive

In this method, the synchronous motor is supplied by a variable frequency inverter in an open loop. Open loop means there is no feedback to the supply, so the inverter doesn't know the rotor's position. This method is suitable when precise speed control isn't necessary. The main supply is fed into a rectifier inverter set to achieve the desired frequency. The motor's synchronous speed can be varied according to this frequency.



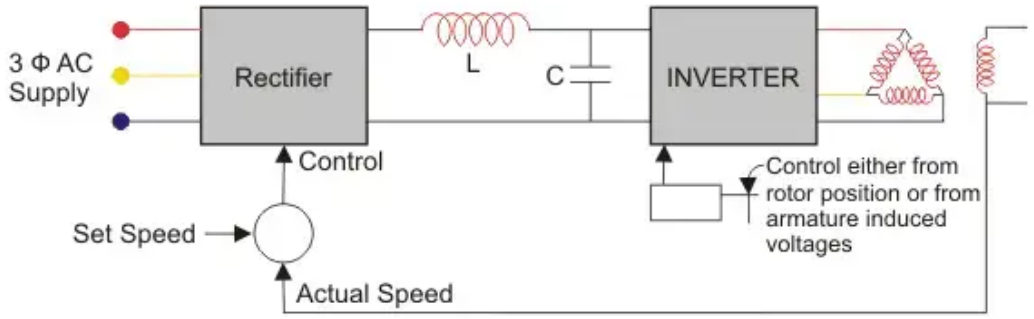


Separate Control of Synchronous Motor Fed from Square Wave Inverter

In above figure, you can see the block diagram of the speed control drive. The three-phase supply from the mains is converted to dc by using rectifiers. Then the rippled dc is smoothened by using LC filters. The DC is fed to the inverters. These inverters can be either voltage source inverters or current source inverter. The frequency of the supply fed to the motor can be varied and accordingly speed control of synchronous motor can be done. Open loop operation is useful when a number of motors need to be run at exactly the same speed. This method has a disadvantage though. This method causes spontaneous oscillation or hunting.

Self Synchronous (Closed – Loop) Operation

Self-synchronous (closed-loop) operation is used for precise speed control. Here, the inverter output frequency is based on the rotor speed. The rotor speed is fed back to a differentiator, and the difference between preset and actual speed is sent to the rectifier. The inverter then adjusts the frequency to control the motor speed accurately.



Self-Control of Synchronous Motor Fed from Square Wave Inverter

For example, if the speed drops due to increased load, the stator supply frequency reduces to keep the rotor in sync with the stator magnetic field. This method prevents spontaneous oscillation or hunting.

Performance with current-source inverter (CSI) drive

The scheme of figure 4.4.2 is used in high power synchronous motor drives. A variable DC current source is established by closed loop control of a DC power supply. The phase angle controlled AC-DC thyristor bridge converter terminated with a large DC link inductor serves as the regulated but stiff current source. The scheme is suitable for large synchronous motors for which thyristor switches are used in the inverter. The current loop is established by sensing the DC link current and by using a closed loop current controller which continuously regulates the firing angle of the front-end controllable AC-DC rectifier. Switching of the six thyristors are synchronised with the phase back emf waveforms, the zero crossings of which are obtained from three Hall sensors (H) mounted on the shaft. It will be shown later that with this type of control, the motor torque is proportional to level of the DC link current.

Note that the dotted trace of figure 4.4.3 represents the fundamental component of the quasi-square current of phase '*a*'. Switching of the six thyristors in converter driving the motor takes place autonomously, arranged by position sensor *E* and the converter switching logic. The angular displacement γ of the phase current waveform (or its fundamental component) with the respect to the back emf waveform of the corresponding phase is indicated in figure 4.4.1. Due to the presence of the large DC-link inductor, phase currents may be considered to remain essentially constant between switching intervals. The quasi-square phase current waveforms contain many harmonics, which may be responsible for some torque pulsations for this drive at low speed.

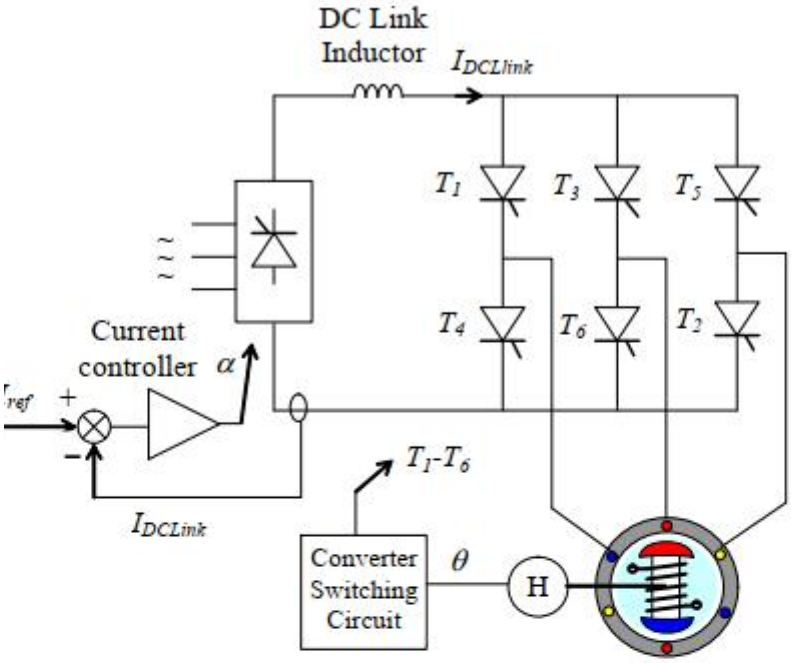


Figure 4.4.2 CSI synchronous motor drive with DC link current control

The above scheme uses a thyristor converter for driving the motor. The motor is usually operated with over excitation, so that the thyristor switches can commutate with the help of the motor back emf for turning the thyristors off at the end of a conduction period. Note that with over-excitation, the motor current leads the back emf so that when a phase current passes through zero, the anode-cathode voltage across the conducting thyristor is favourable i.e., the out going thyristor is reverse biased by the the back emf of the motor. This type of drive is normally used in very large power applications for which the only suitably rated, natuarally commutated thyristor switches are available..

The motor can be reversed easily by reversing the phase sequence of switching of the inverter. The drive can also be braked regeneratively by increasing the firing angle of the input rectifier above 90° while maintaining the DC link current at the desired braking torque level until braking is no longer required. The rectifier now absorbs the energy of the overhauling motor, regeneratively.

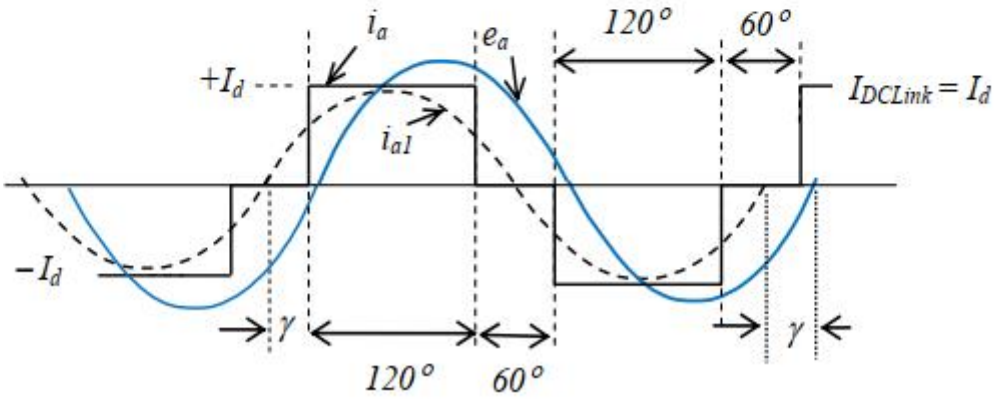
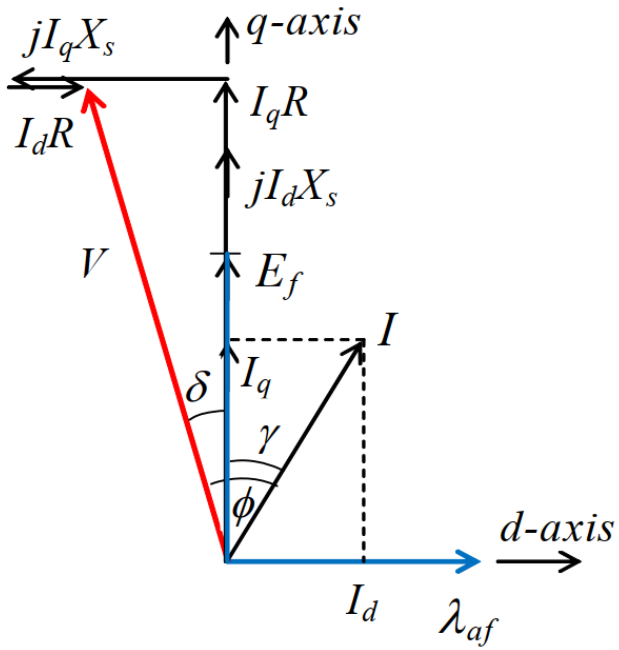


Figure 4.4.3

Brushless DC operation of CSI driven synchronous motor

In the following analysis, it is assumed that the supply current waveform for each phase is sinusoidal and of controllable amplitude or RMS value. It is also assumed that the phase angle between current and induced back emf in each phase can be arbitrarily chosen. All of these tasks are arranged through continuous rotor position feedback and continuous control of stator currents in closed loops. In other words, phase current references, and hence actual rotor currents, are assumed to be synchronised with the rotor position (angle). [In a more general sense, this type of control is equivalent to controlling the motor currents in the rotor reference frame].



Phasor diagram

Unity power factor operation

The power factor with which a motor operates is an important issue, especially for a large drive. The power factor is given by the cosine of the angle ϕ between the input voltage and current to the motor. A large ϕ results in a poor power factor which means that in order to deliver a load, the motor must draw a larger current than what would be drawn from the inverter if the power factor were high. For a large drive, a poor power factor may not be acceptable from the consideration of the cost associated with supply of high input current from the supply.

Operation of the synchronous motor with a current source inverter allows power factor compensation directly, using the commutation angle (γ) adjustment. This is more direct than via excitation control for voltage source inverter drives. Consider the following two cases.

I lags E_f

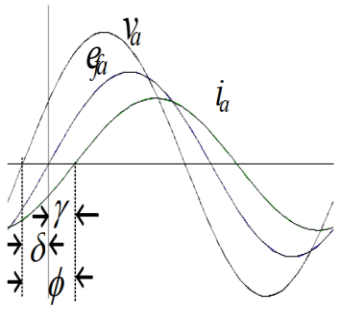


Figure 4.4.7(a)

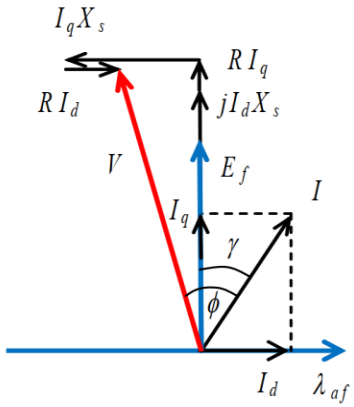


Figure 4.4.7(b)

In Figure 4.4.7, the overall power factor is lagging, since γ is a lagging angle. The power factor angle ϕ is larger than γ. Note that since the motor is under-excited and I lags E_f, and I_d *magnetizes* the rotor.

I leads E_f

If the I phasor is chosen to lead E_f, the overall power factor can be much better, as indicated in Figure 4.4.8, including unity. Note that the motor now operates with less than maximum torque per Ampere (MTPA) characteristic.

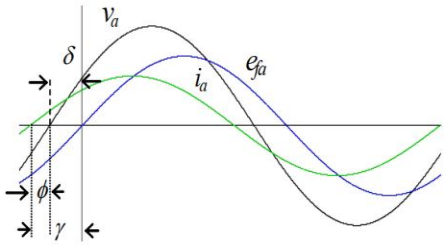


Figure 4.4.8(a)

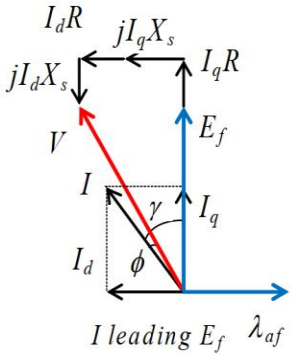


Figure 4.4.8(b)